

Final Project Report

Smartphone Usage, Digital Addiction & Well-Being Analytics

1. Introduction

1.1 Project Overview

Smartphone Usage, Digital Addiction & Well-Being Analytics is a Power BI-based data analytics project developed to analyze smartphone usage patterns and their relationship with digital addiction and user well-being.

The project uses **7,500 user records with 16 attributes**, including demographic information, daily smartphone usage, social media and gaming usage, notifications, app openings, sleep, stress, academic/work impact, and addiction classification.

A major finding from the dataset is that **5,308 out of 7,500 users (70.77%) are classified as addicted**. The project therefore focuses on understanding the usage, lifestyle, stress, and demographic factors associated with this classification.

Power BI is used to clean, transform, analyze, and visualize the data through an interactive dashboard.

1.2 Objectives

The main objectives of the project are:

- Analyze overall smartphone addiction prevalence.
- Compare addiction across age groups and gender.
- Analyze screen-time and smartphone usage behavior.
- Examine social media and gaming usage.
- Analyze notifications and app-opening frequency.
- Study the relationship between smartphone usage, sleep, and stress.
- Analyze reported academic/work impact.
- Create an **Age Group** calculated field for segment-level analysis.
- Develop an interactive Power BI dashboard.
- Provide clear insights to support digital well-being analysis.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The organization needs to understand why **5,308 of its 7,500 users (70.77%)** are classified as addicted to smartphone usage.

The analysis focuses on determining which factors correspond with the addiction classification, including:

- Screen-time behavior
- Social media usage
- Gaming
- Notifications
- App openings
- Sleep
- Stress
- Age
- Gender
- Academic/work impact

Without proper analysis, product and research teams may have limited visibility into user behavior and digital well-being patterns.

Therefore, the project aims to provide a centralized Power BI dashboard for exploring these factors and identifying meaningful patterns in the available data.

2.2 Project Proposal — Proposed Solution

The proposed solution is to develop an **interactive Power BI dashboard** that combines smartphone usage, demographic, lifestyle, stress, and addiction information.

The solution includes:

1. Importing the Smartphone Usage dataset.
2. Cleaning and preparing the data using Power Query.
3. Creating required calculated fields such as Age Group.
4. Analyzing addiction classification.
5. Creating KPIs and visualizations.
6. Adding interactive filters and slicers.
7. Presenting important findings through a Power BI dashboard.
8. Using the dashboard to support further digital well-being analysis.

The proposed solution provides an easy-to-understand visual representation of the dataset instead of relying on manual analysis of individual records.

2.3 Initial Project Planning

Phase	Activities
Project Initialization	Define project topic and business problem
Data Collection	Collect Smartphone_Usage.csv
Data Understanding	Review columns, data types, and categories
Data Preprocessing	Clean and transform the dataset
Data Analysis	Analyze usage and addiction patterns
Calculated Fields	Create Age Group and required measures
Visualization	Develop charts, cards, and tables
Dashboard Development	Create interactive Power BI dashboard
Testing	Test filters, calculations, and visuals
Reporting	Document findings and observations
Finalization	Review dashboard and final report

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Dataset Name: **Smartphone_Usage.csv**

Dataset Size

- **Records:** 7,500
- **Attributes:** 16

The dataset contains information related to smartphone usage behavior, demographics, lifestyle, stress, and addiction.

Main Data Categories

Demographics

- Age
- Gender

Usage Behavior

- Daily screen time
- Social media hours
- Gaming hours
- Work/study hours
- Weekend screen time

Smartphone Engagement

- Notifications per day
- App opens per day

Well-Being

- Sleep hours
- Stress level
- Academic/work impact

Addiction

- Addiction level
- Addicted label

3.2 Data Quality Report

The dataset was reviewed before analysis to ensure that it was suitable for Power BI reporting.

The data quality process focused on:

- Checking column names.
- Verifying data types.
- Checking missing values.
- Checking duplicate records.
- Reviewing categorical values.
- Checking numerical fields.
- Ensuring the addiction label is suitable for analysis.

The target field is:

Addicted label

where:

- 1 = Addicted
- 0 = Not Addicted

The provided project information identifies 5,308 addicted users out of 7,500 records.

3.3 Data Exploration and Preprocessing

The data was prepared using Power Query in Power BI.

The preprocessing activities include:

Data Import

The Smartphone_Usage.csv dataset was imported into Power BI.

Data Type Checking

Numerical fields such as age, screen time, sleep hours, notifications, and app openings were prepared for numerical analysis.

Categorical fields such as gender, stress level, addiction level, and academic/work impact were prepared for category-based analysis.

Data Transformation

The dataset was transformed to make it suitable for analysis and visualization.

Age Group

An Age Group calculated field was created to support demographic segmentation and make age-based analysis easier.

4. Data Visualization

4.1 Framing Business Questions

The Power BI analysis focuses on questions such as:

1. How many users are classified as addicted?
2. What percentage of users are addicted?
3. How does addiction vary by gender?
4. How does addiction vary across age groups?
5. How does screen time relate to addiction classification?
6. How does social media usage vary among users?

7. How does gaming usage vary across addiction categories?
8. How do notifications and app openings differ?
9. How does sleep duration vary across addiction classifications?
10. How does addiction classification vary by stress level?
11. How does smartphone usage affect academic/work performance?
12. What patterns can be observed between usage behavior and addiction classification?

4.2 Developing Visualizations

The Power BI report can include the following visualizations:

Visualization	Purpose
Total Users KPI	Shows total number of users
Addicted Users KPI	Shows number of addicted users
Addiction Percentage KPI	Shows overall addiction percentage
Addiction by Gender	Compares addiction across gender
Addiction by Age Group	Compares different age segments
Addiction by Stress Level	Analyzes stress categories
Addiction Level Chart	Shows addiction severity
Screen Time Analysis	Examines daily screen usage
Social Media Analysis	Examines social media usage
Gaming Analysis	Examines gaming usage
Sleep Analysis	Examines sleep duration
Notifications Analysis	Examines notification frequency
App Opens Analysis	Examines interaction frequency
Academic/Work Impact	Shows reported impact

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

The Power BI dashboard provides a consolidated view of smartphone usage, addiction, and well-being indicators.

Important Dashboard KPIs

Total Users: 7,500

Addicted Users: 5,308

Addiction Percentage: 70.77%

Not Addicted Users: 2,192

Dashboard Insights

- A significant proportion of the analyzed users are classified as addicted.
- Screen-time behavior can be examined alongside addiction classification.
- Social media and gaming provide additional usage dimensions.
- Stress and sleep provide important well-being indicators.
- Age Group allows easier demographic comparison.
- Interactive filters allow users to explore individual segments.

The dashboard is designed to provide a simple and interactive way to explore the available data.

6. Report

6.1 Report Design Screenshot with Insights

The Power BI report presents detailed analysis of smartphone usage and digital addiction.

The report can contain multiple pages or sections covering:

Overview

Provides major KPIs and overall addiction information.

Demographic Analysis

Analyzes:

- Age
- Age Group
- Gender

Usage Analysis

Analyzes:

- Screen time
- Social media
- Gaming
- Work/study usage
- Weekend usage

Well-Being Analysis

Analyzes:

- Sleep
- Stress
- Academic/work impact

Addiction Analysis

Analyzes:

- Addicted vs Not Addicted
- Addiction levels
- Addiction percentage by segment

The report allows users to interact with visualizations and filters to explore the data.

7. Performance Testing

7.1 Utilization of Data Filters

Interactive filters/slicers are used to analyze specific segments of the dataset.

Possible filters include:

- Gender
- Age Group
- Stress Level
- Addiction Level
- Academic/Work Impact
- Addicted Label

These filters allow users to dynamically change the displayed analysis.

7.2 Number of Calculation Columns/Measures

The project uses calculated fields and measures required for dashboard analysis.

A major calculated field is:

Age Group

The Age Group field categorizes users into meaningful age segments.

Additional measures can be used for:

- Total Users
- Addicted Users
- Not Addicted Users
- Addiction Percentage
- Average Screen Time
- Average Sleep Hours
- Average Social Media Hours

Note: The exact number of calculated columns/measures should be entered according to the final .pbix file if your template requires an exact count.

7.3 Number of Visualizations

The dashboard/report can contain multiple visualizations including:

- KPI cards
- Bar charts
- Column charts
- Pie/donut charts
- Tables
- Slicers
- Other Power BI visuals used in the final dashboard

Note: The exact number should be entered based on the final Power BI dashboard rather than estimated.

8. Conclusion / Observation

The **Smartphone Usage, Digital Addiction & Well-Being Analytics** project demonstrates how Power BI can be used to analyze smartphone usage and digital addiction patterns.

The dataset contains **7,500 users and 16 attributes**, providing information about demographics, smartphone behavior, lifestyle, stress, and addiction.

A key observation is that **5,308 users (70.77%) are classified as addicted**, making addiction classification an important area for further analysis.

The project examines multiple factors including screen time, social media usage, gaming, notifications, app openings, sleep, stress, and academic/work impact.

The Power BI dashboard provides an interactive environment where users can filter the data and examine different segments. The **Age Group** calculated field also improves demographic analysis.

Overall, the project converts raw smartphone usage data into a structured analytical report that can support further research into digital well-being and healthier smartphone usage patterns.

9. Future Scope

The project can be enhanced in the future by:

- Adding machine learning-based addiction prediction.
- Adding historical/time-series smartphone usage analysis.
- Including additional well-being variables.
- Integrating real-time smartphone usage data.
- Creating personalized digital well-being insights.
- Adding more detailed app-category analysis.
- Developing automated reports.
- Publishing the dashboard through Power BI Service.
- Adding predictive analytics for future usage patterns.

10. Appendix

- **10.1 Source Code**

The project is primarily developed using **Power BI and Power Query**.

Relevant Power Query transformations and Power BI calculated fields/measures can be included here if required.

- **10.2 GitHub & Project Demo Link**

GitHub Repository: <https://github.com/Usharani18/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>

Project Summary

Item	Details
Project Name	Smartphone Usage, Digital Addiction & Well-Being Analytics
Tool Used	Microsoft Power BI
Data Source	Smartphone_Usage.csv
Records	7,500
Attributes	16
Addicted Users	5,308
Addiction Percentage	70.77%
Target Variable	addicted_label
Main Calculated Field	Age Group
Main Analysis	Usage, addiction, demographics & well-being
Dashboard Type	Interactive Power BI Dashboard
Primary Purpose	Analyze smartphone usage and digital addiction patterns